COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	7/7/26	**DATE SUBMITTED:**	7/9/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm:

Get three grade values:

Input grade1

Input grade2

Input grade3

total= Add grade1, grade2, and grade3

Average = Divide total by 3

Print Average

Pseudocode:

Start

Input three grades

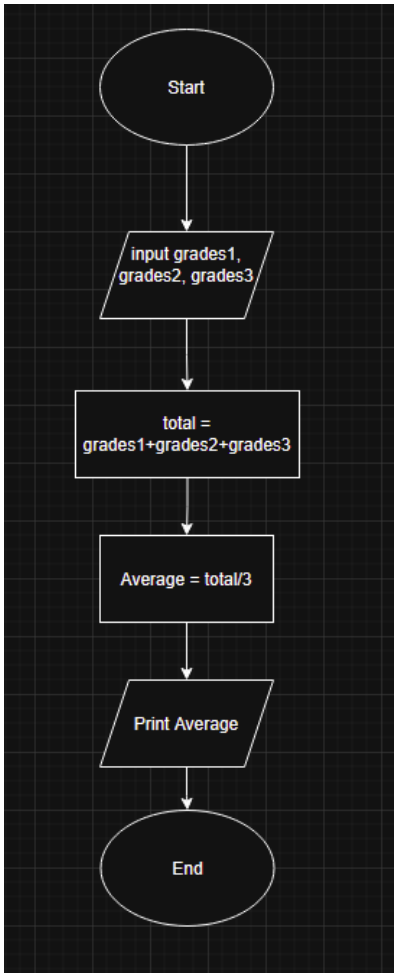
Add the value of grades

Divide the sum by 3

Print the average of grades

End

Flowchart:


Start
input grades1, grades2, grades3
total = grades1+grades2+grades3
Average = total/3
Print Average
End

Code:

Seatwork.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {

    float grade1 = 0;
    float grade2 = 0;
    float grade3 = 0;
    float sum = 0;
    float Average = 0;

    cout<< "Input your grade: "<< endl;
    cin >> grade1;

    cout << "Input your grade: "<< endl;
    cin>> grade2;

    cout << "Input your grade: " << endl;
    cin>> grade3;

    sum = grade1 + grade2 + grade3;
    Average = sum/3;

    cout << "Your average grade is: " << Average << endl;


    return 0;
}
```

```
PS C:\Users\TIPQC\OneDrive\Documents\Urbano-C++> C:\Users\TIPQC\OneDrive\Documents\Urbano-C++\Seatwork.exe
Input your grade:
87
Input your grade:
88
Input your grade:
90
Your average grade is: 88.3333
PS C:\Users\TIPQC\OneDrive\Documents\Urbano-C++>
```

Code Analysis:

In this code, we can see that I have deployed a float named grade1,grade2 and grade3. These will be responsible for storing the grades that the user will input. After the user inputs the grades, it will add the grades and get the average which will then be printed.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm:

Get three prices of goods:

    Input prices

Total = Add the prices

Print transaction

Print total

Pseudocode:

Start

Get the values from the user

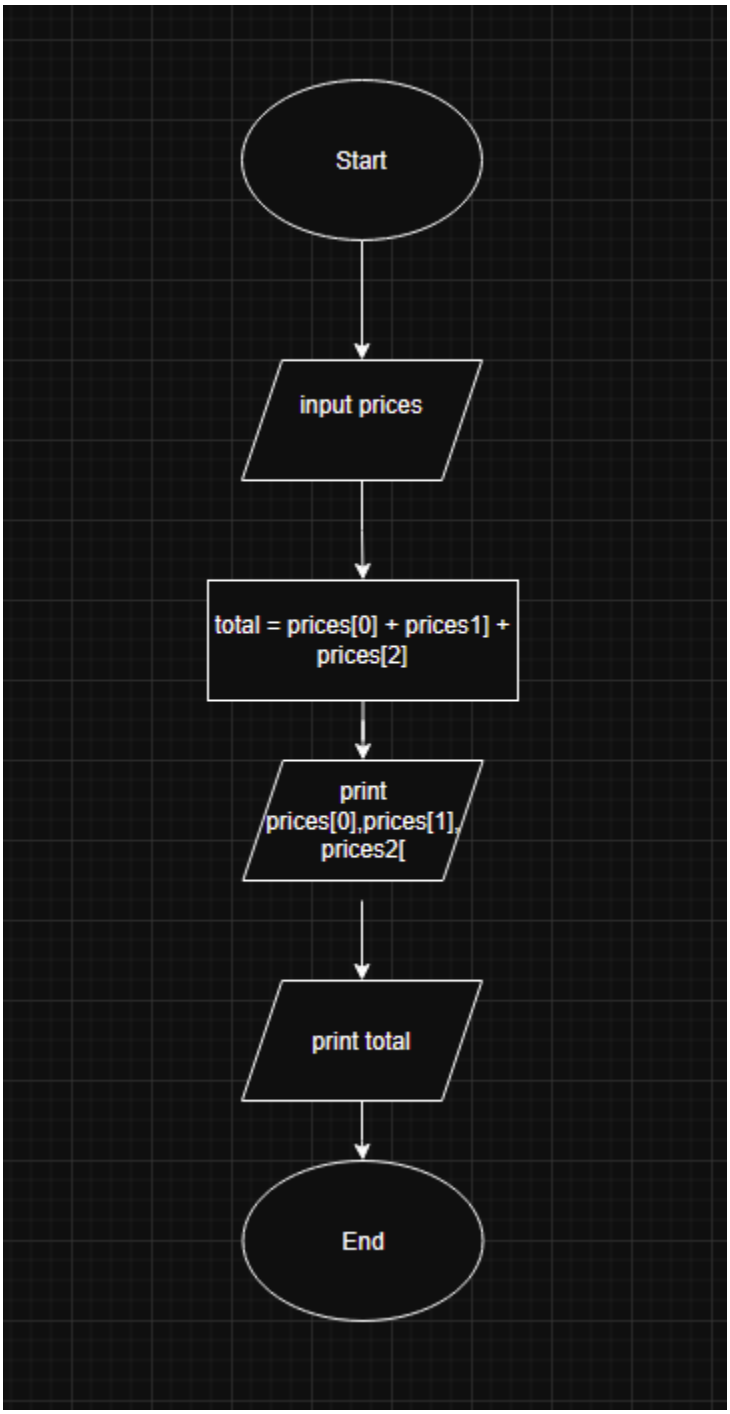
Add the values

Print the sum as total

Print the transaction History

End

Flowchart:


Start
input prices
total = prices[0] + prices1] + prices[2]
print prices[0],prices[1], prices2[
print total
End

```cpp
#include <iostream>
using namespace std;
int main() {
    int goods = 3;
    double price[goods];
    double total = 0;

    for (int i = 0; i < goods; i++){
        cout << "Please enter the price of your goods: ";
        cin>> price[i];
        {
            total = price[0] + price[1] + price[2];

        }
    }
    cout<< "Transaction History: \n"<<"First goods: " <<price[0] <<"\nSecond Goods: "<< price[1] << "\nThird Goods: "<< price[2]<<endl;
    cout<< "Total: " <<total<< endl;
    return 0;
}
```

```
PS C:\Users\TIPQC\Documents\C++> C:\Users\TIPQC\Documents\C++\Seatwork2.exe
Please enter the price of your goods: 46
Please enter the price of your goods: 26.5
Please enter the price of your goods: 24.8
Transaction History:
First goods: 46
Second Goods: 26.5
Third Goods: 24.8
Total: 97.3
```

Code analysis:

In this code, we have used an array to store the prices of the goods. The arrays will be responsible for printing the transaction history. After getting the prices, the prices will be added and will be stored to the total. It will all be printed in the transaction history.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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